Time Series Forecasting Project

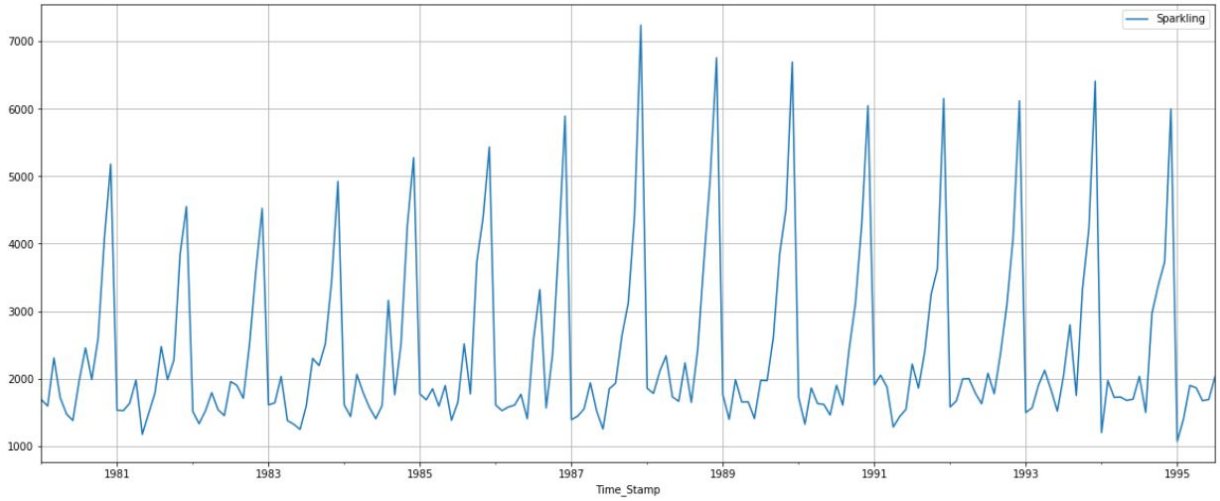
Problem 1

For this particular assignment, the data of different types of wine sales in the 20th century is to be analysed. Both of these data are from the same company but of different wines. As an analyst in the ABC Estate Wines, you are tasked to analyse and forecast Wine Sales in the 20th century.

1. Read the data as an appropriate Time Series data and plot the data.

Sparkling

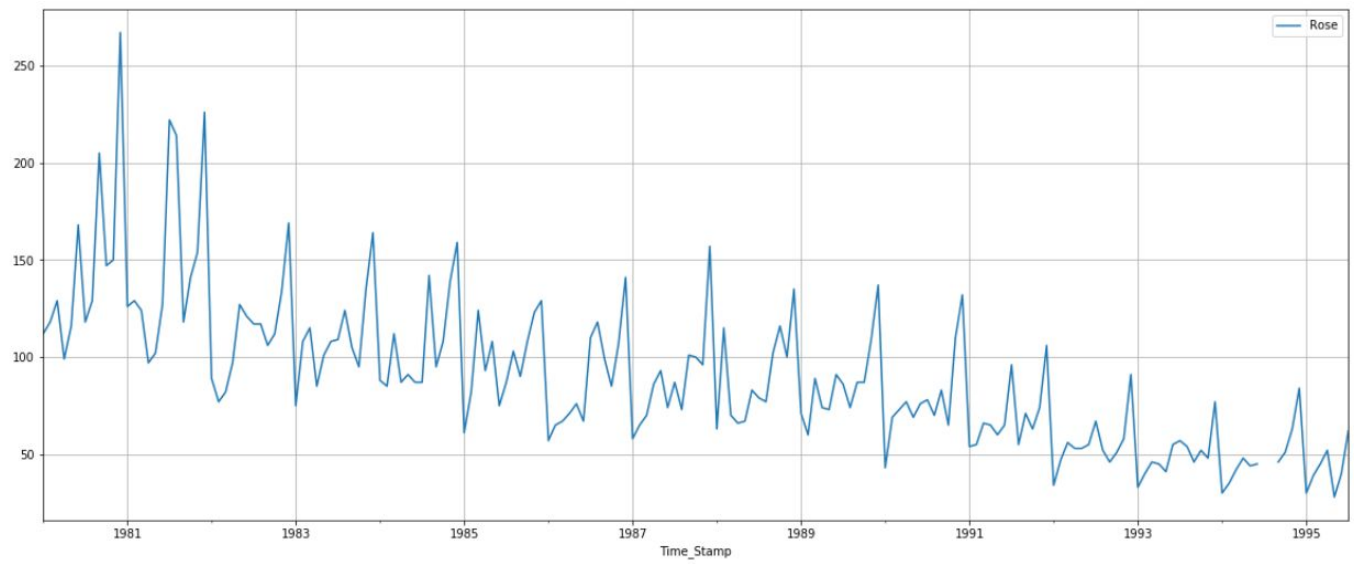
Time_Stamp	Sparkling
1980-01-31	1686
1980-02-29	1591
1980-03-31	2304
1980-04-30	1712
1980-05-31	1471



- The Dataset has been read as a Time series data and been plotted
- The data begins in January 1980 and ends in July 1995
- The Data is a Monthly series
- The plot shows Multiplicative Seasonality with little to no Trend

Rose

Time_Stamp	Rose
1980-01-31	112.0
1980-02-29	118.0
1980-03-31	129.0
1980-04-30	99.0
1980-05-31	116.0



- The Dataset has been read as a Time series data and been plotted
- The data begins in January 1980 and ends in July 1995
- The Data is a Monthly series
- The plot shows inverse Multiplicative Seasonality with slightly decreasing trend
- The plot shows missing values at 1994 which has been imputed using Interpolate Function

2. Perform appropriate Exploratory Data Analysis to understand the data and also perform decomposition.

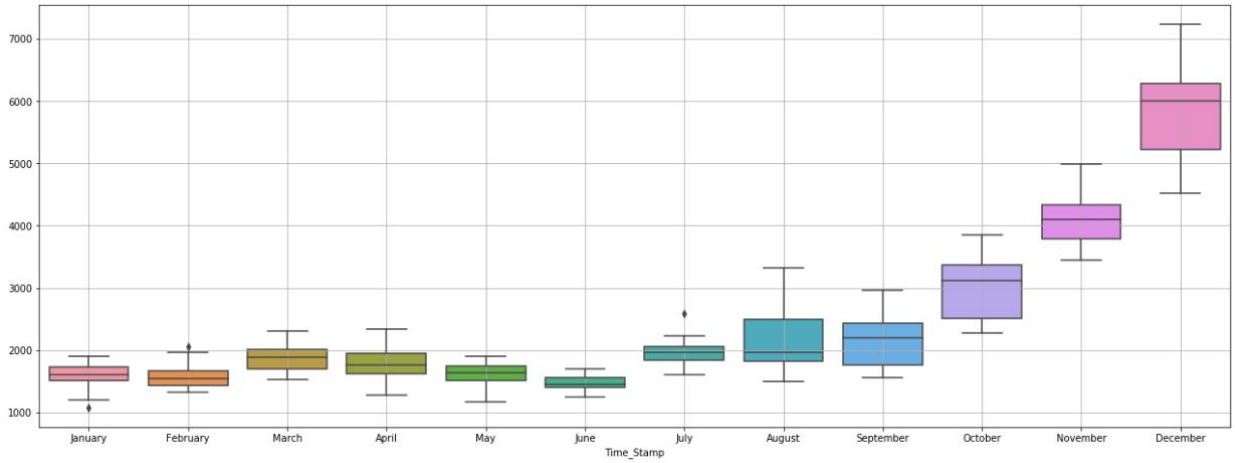
Sparkling

Description the Data

	Sparkling
count	187.000000
mean	2402.417112
std	1295.111540
min	1070.000000
25%	1605.000000
50%	1874.000000
75%	2549.000000
max	7242.000000

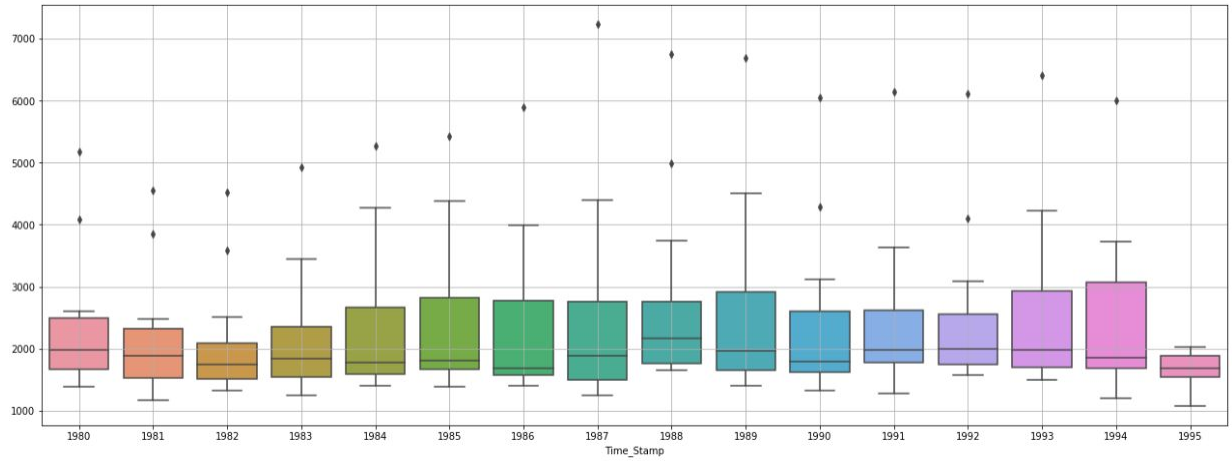
- The Average Sales for Sparkling Wine from 1980-1995 is 2402
- The minimum sales is 1070 and maximum sales is 7242

Monthly Boxplot



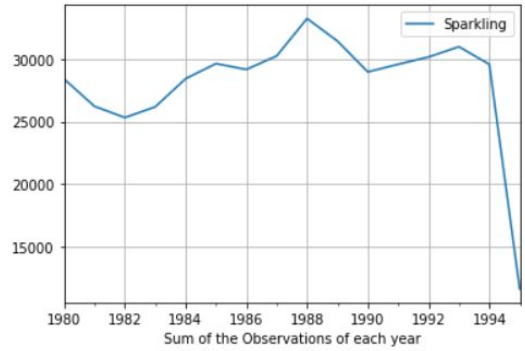
- Wine Sales is low in the first half of the year and gradually increases from July
- From October there is a rapid hike in Sales
- December has the highest Wine Sales

Yearly Boxplot



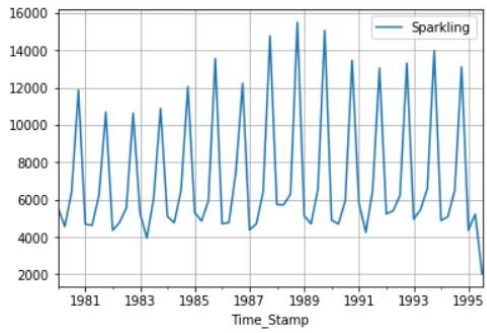
- 1987 has the highest Wine Sales
- 1995 has the lowest Wine Sales
- All the Years have a roughly equal sales except for 1995

Resampled Yearly plot



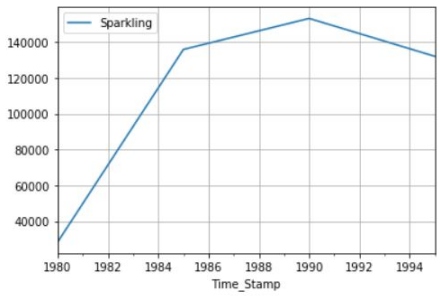
- The Trend is Visible in this resampled yearly plot
- There is a dip in 1982 and gradual hike after that till 1988
- After 1988 sales fall after a slight increase

Resampled Quarterly Plot



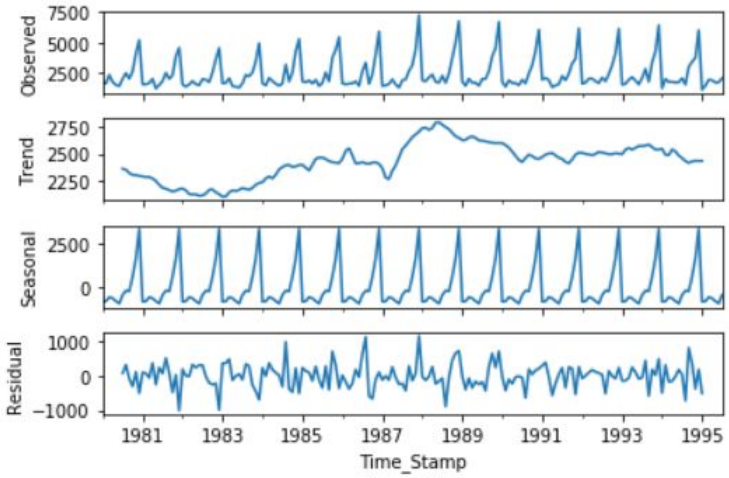
- The seasonality is visible in this resampled quarterly plot
- The seasonality increases from 1983 which is a sign of Multiplicative behaviour

5 Year Plot



- The 5 Year plot shows behaviour of an increasing trend 1985

Decomposition



- The dataset has been decomposed
- The seasonality and Trend is very visible now as they have been separated

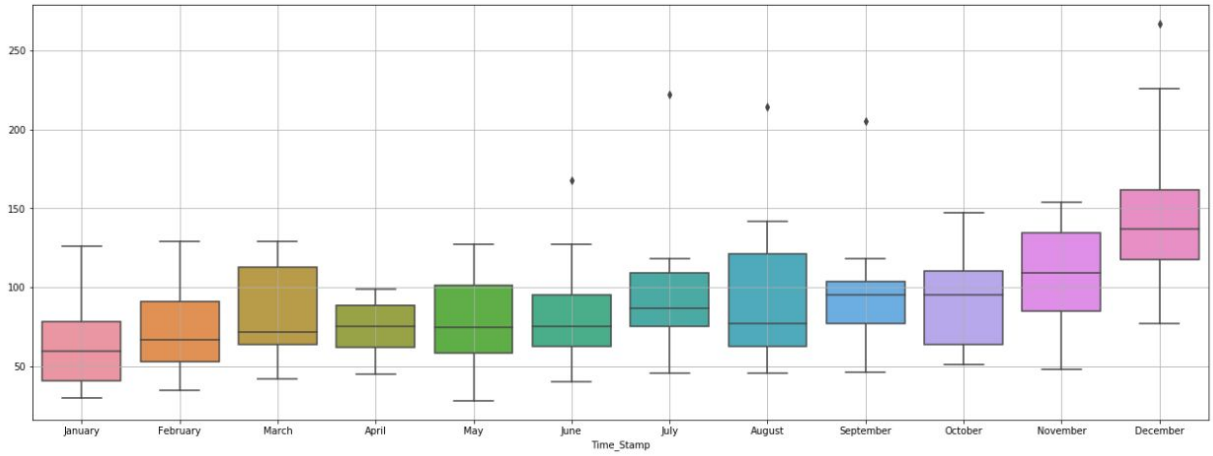
Rose

Description of the Data

	Rose
count	187.000000
mean	89.914439
std	39.238325
min	28.000000
25%	62.500000
50%	85.000000
75%	111.000000
max	267.000000

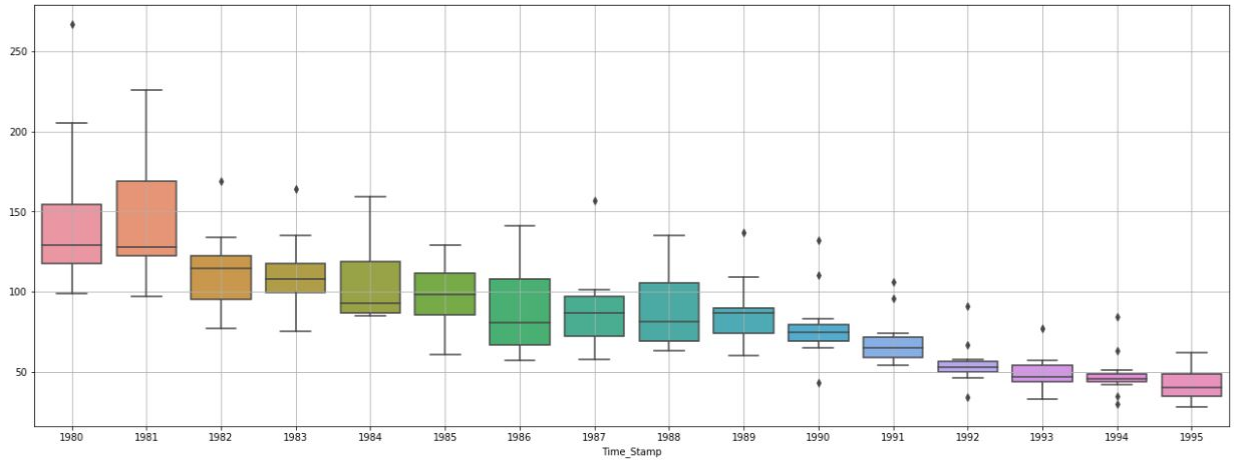
- The Average Sales for Rose Wine from 1980-1995 is 89.9
- The minimum sales is 28 and maximum sales is 267

Monthly Boxplot



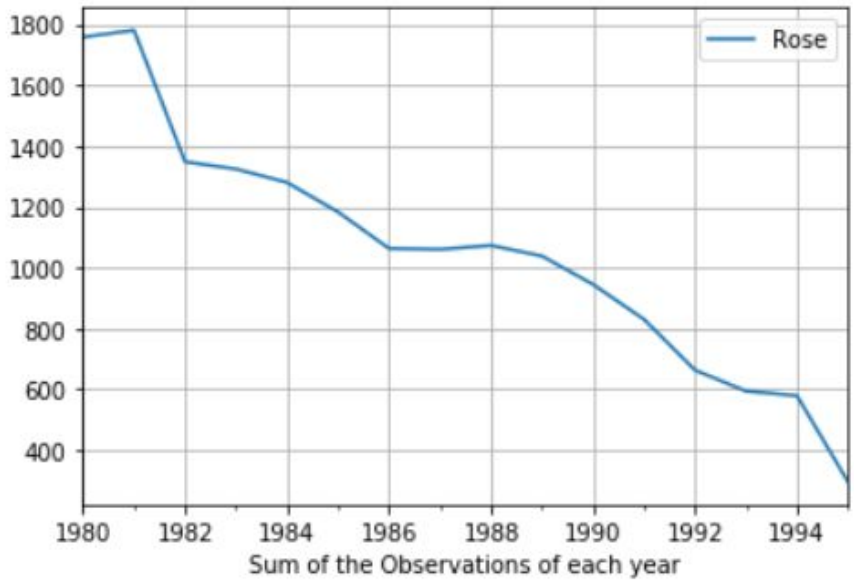
- Wine Sales is low in the first half of the year and gradually increases from July
- December has the highest Wine Sales

Yearly Boxplot



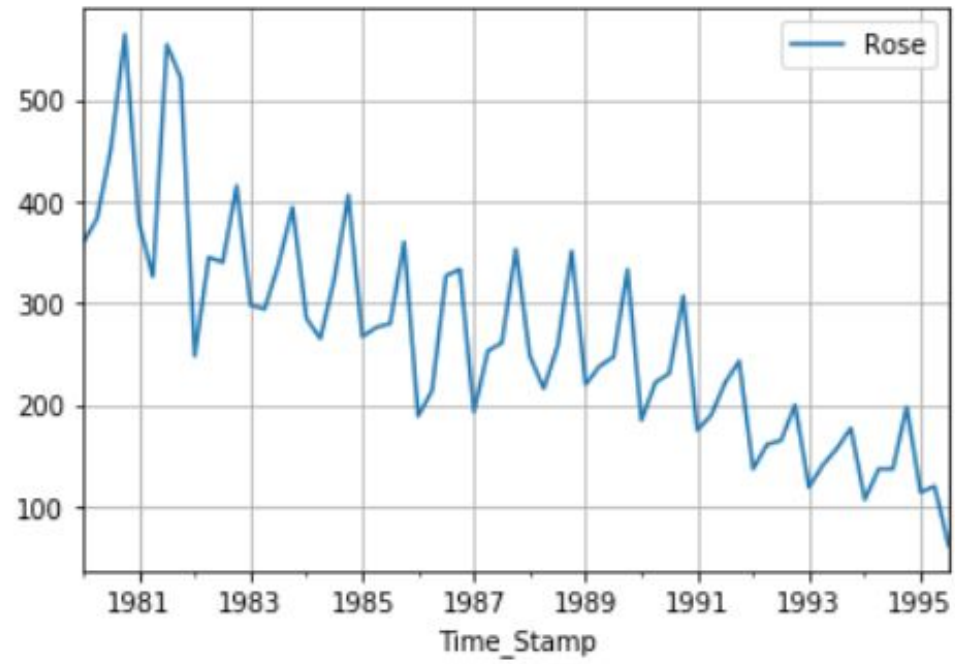
- 1980 has the highest Wine Sales
- After 1980 it gradually keeps decreasing till 1995

Resampled Yearly Plot



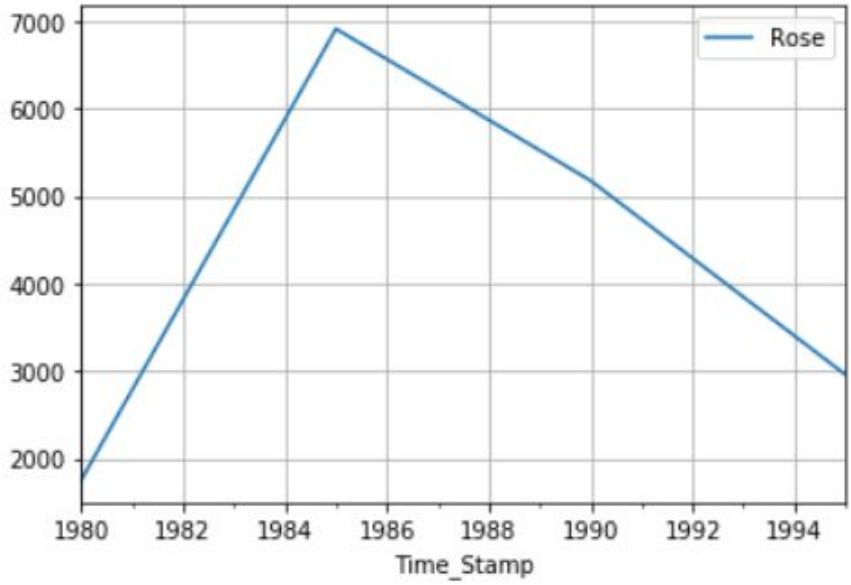
- The resampled yearly plot shows a visible trend which keeps decreasing each year

Resampled Quarterly Plot



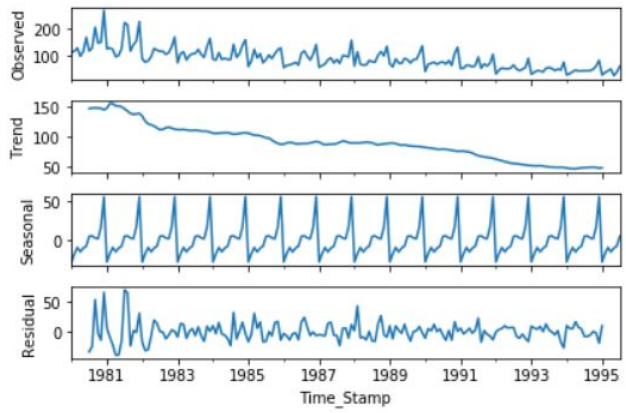
- The Plot shows Multiplicative seasonality of the decreasing kind

5 Year Plot



- The 5 Year Plot shows a hike till 1985 after which it falls drastically till 1995

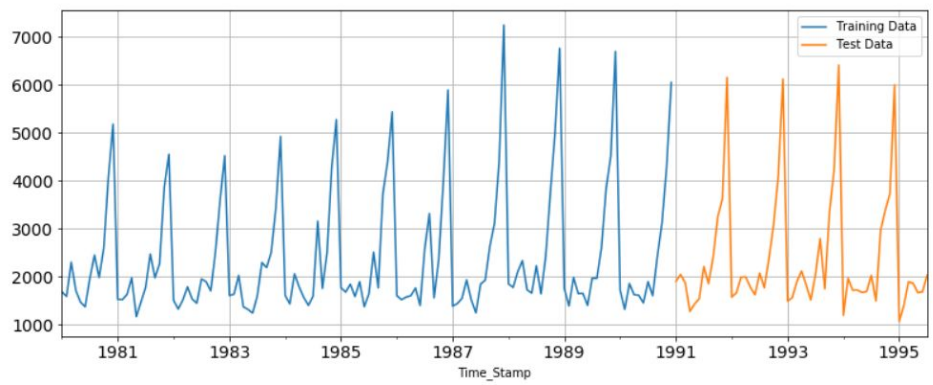
Decomposition



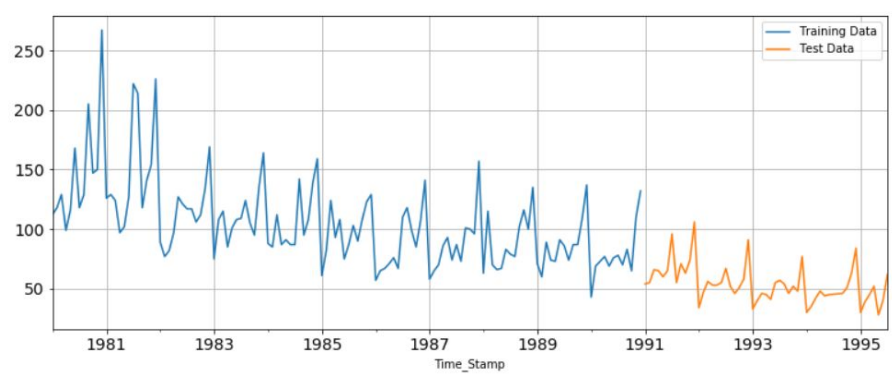
- The decomposition shows trend and seasonality after separation

3. Split the data into training and test. The test data should start in 1991.

Sparkling



Rose

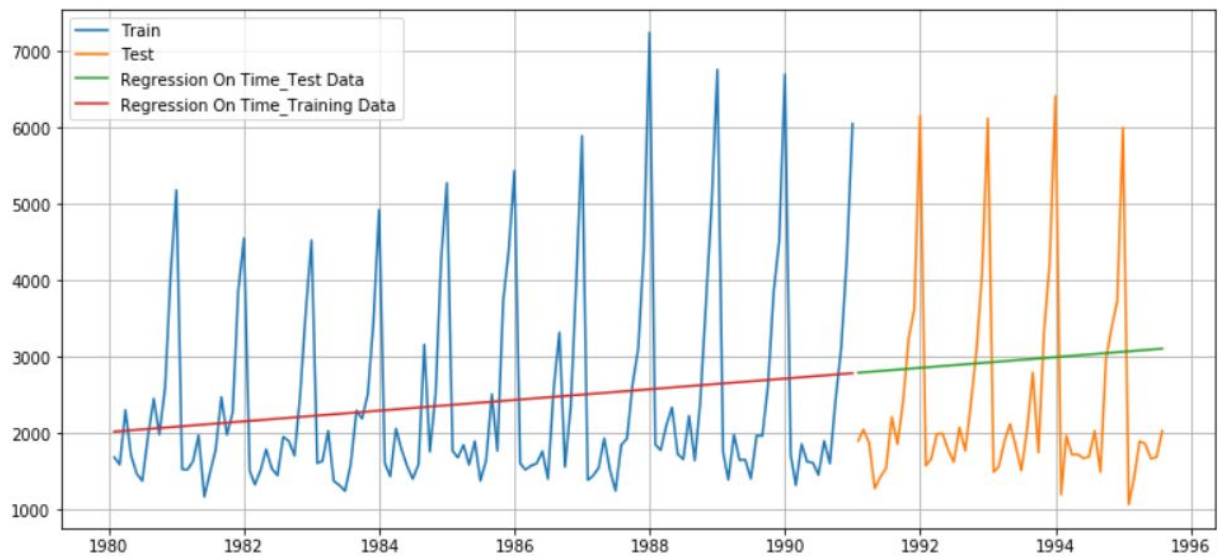


- The data has been separated in such a manner that the Test set starts at 1991 for both datasets
- Train Data has 132 observations and Test Data has 55 observations for Sparkling and Rose
- The data has been plotted after separation into Train and Test

4. Build various exponential smoothing models on the training data and evaluate the model using RMSE on the test data. Other models such as regression,naïve forecast models, simple average models etc. should also be built on the training data and check the performance on the test data using RMSE. - Please do try to build as many models as possible and as many iterations of models as possible with different parameters.

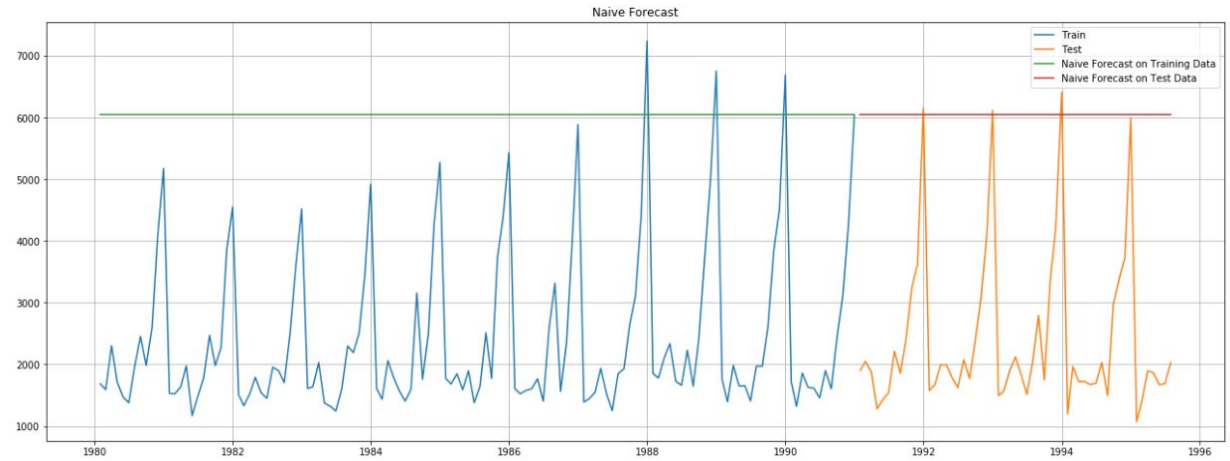
Sparkling

Linear Regression



- Linear Regression model has been fitted and applied on Test Data
- RMSE has been calculated at 1389.135
- The Plot suggests the model fails at predicting the correct sales as it is not able to capture the seasonality of the data

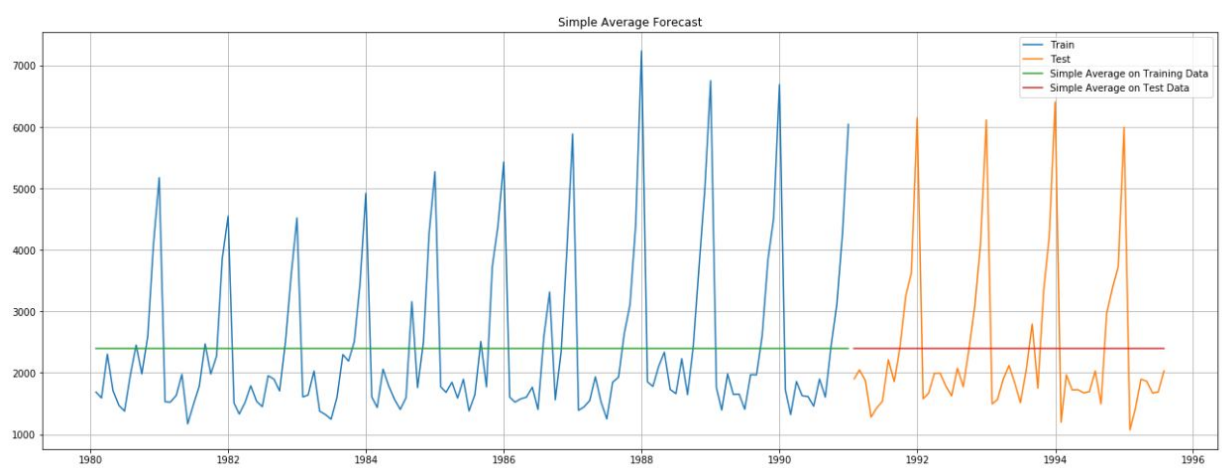
Naive Model



	Test RMSE
RegressionOnTime	1389.135175
NaiveModel	3864.279352

- Naive Model has been built and fitted over Test Data
- The RMSE of the model is higher than that of Linear Regression
- The Plot suggests it has failed to forecast the sales

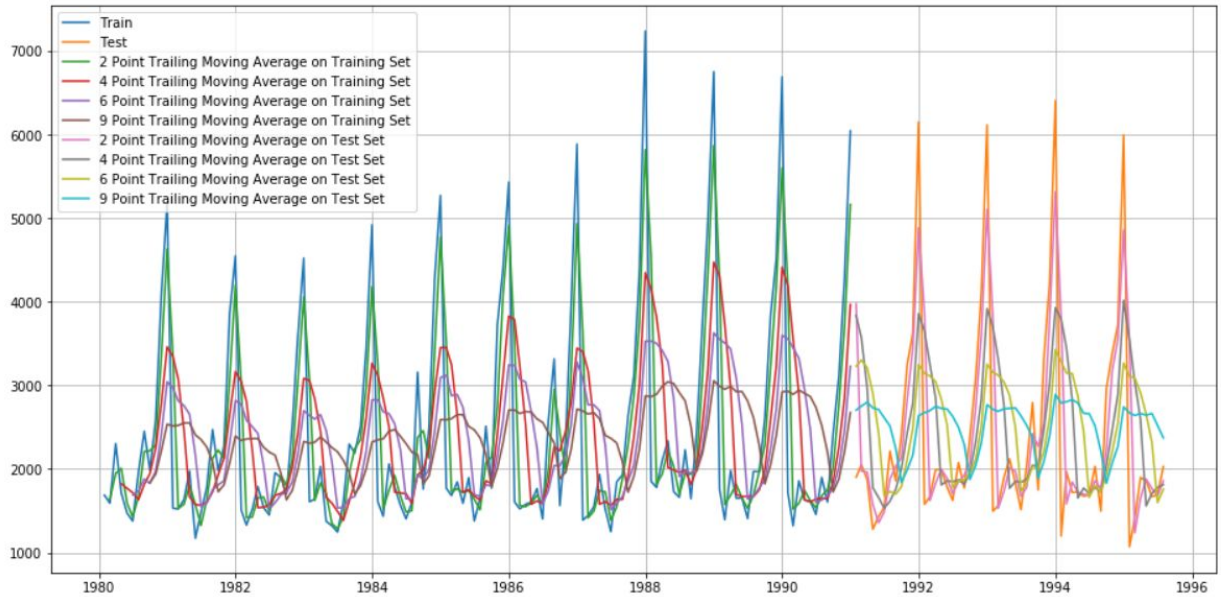
Simple Average



SimpleAverageModel	1275.081804

- The RMSE of the Simple Average Model is lower than the other two models which suggests it has done better at decreasing its errors
- However the plot suggests it too has failed at forecasting correct Sales

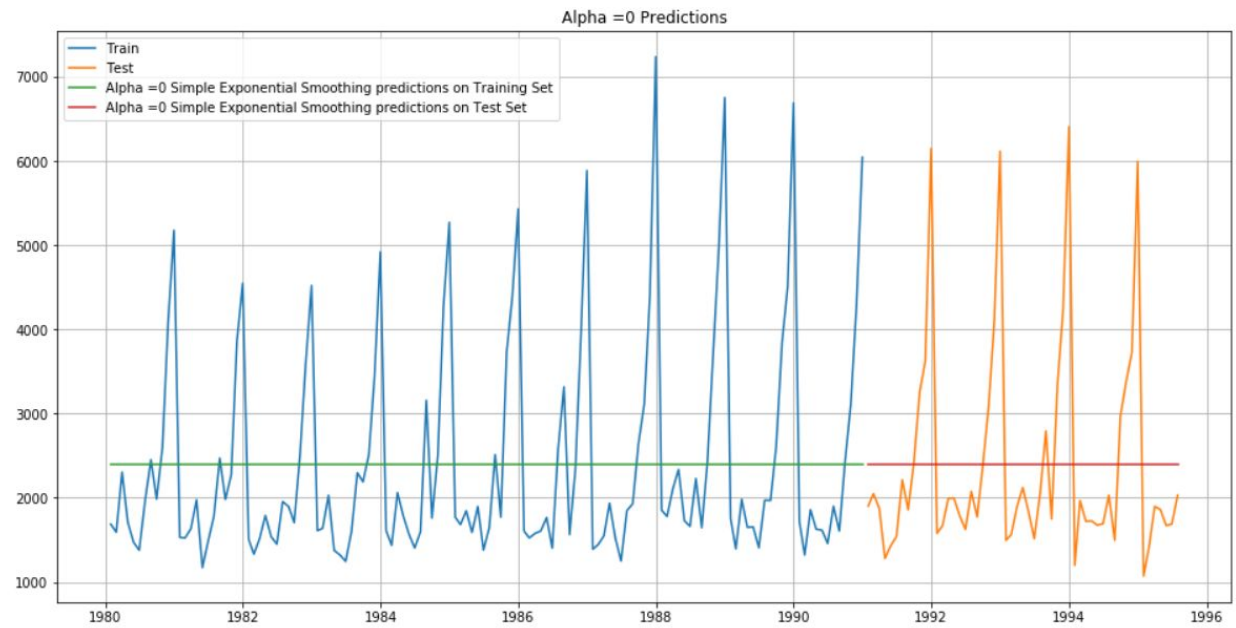
Moving Average



2pointTrailingMovingAverage	813.400684
4pointTrailingMovingAverage	1156.589694
6pointTrailingMovingAverage	1283.927428
9pointTrailingMovingAverage	1346.278315

- Moving Average model has been applied at 2,4,6 and 9 Points
- The RMSE suggests that 2point Moving Average has done a better job at reducing its errors when compared with all other models
- The plot shows that 2point Moving Average comes significantly close to forcasting exact Sales of Sparkling Wine

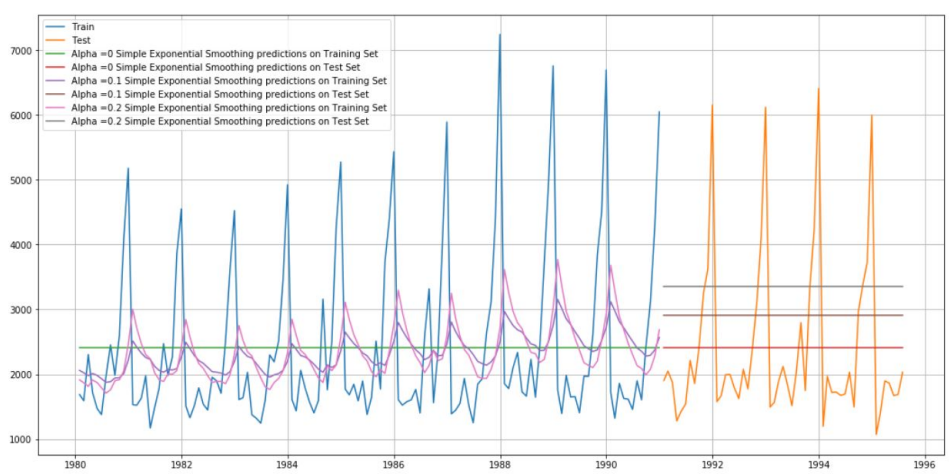
Simple Exponential Smoothing Automated



Alpha=0,SimpleExponentialSmoothing 1275.081823

- Simple Exponential Smoothing Model has been fitted using best parameters suggested by Statsmodels
- At Alpha Level 0 the model's Performance is similar to that of Simple Average according to RMSE
- The plot too looks similar to that of Simple Average

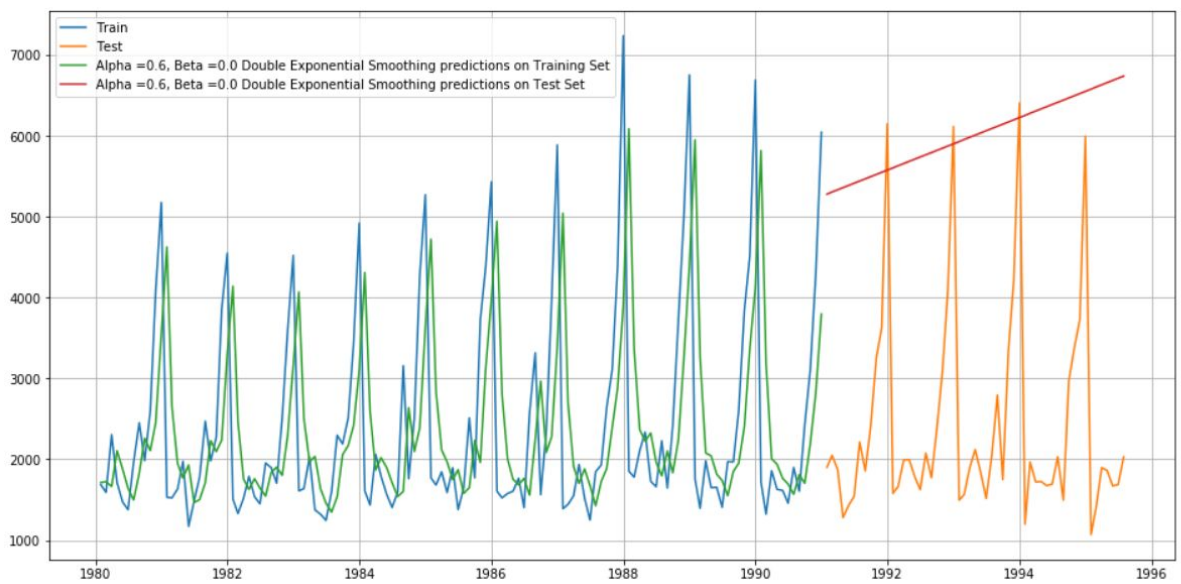
Simple Exponential Smoothing Manual



Alpha=0.1,SimpleExponentialSmoothing	1375.393526
Alpha=0.2,SimpleExponentialSmoothing	1595.206839

- After different iterations of the model it still fails in forecasting correct Sales
- Both at Alpha Level 1 and 2 the model is overfitted as it works on Train but fails on test Data as observed in the plot
- The RMSE of both iterations are high

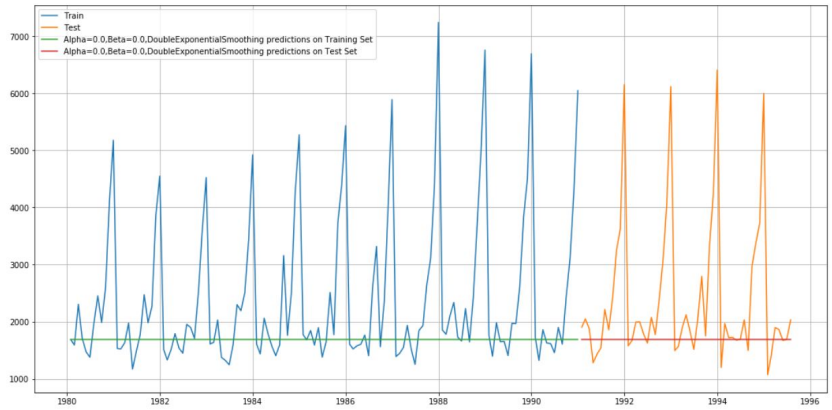
Double Exponential Smoothing Automated



Alpha=0.6 Beta =0,DoubleExponentialSmoothing	3851.171500

- Double exponential smoothing model has been applied using best parameters
- At Alpha Level 0.6 and and Beta level 0 the model's performance is similar to that of Naive Model based on RMSE

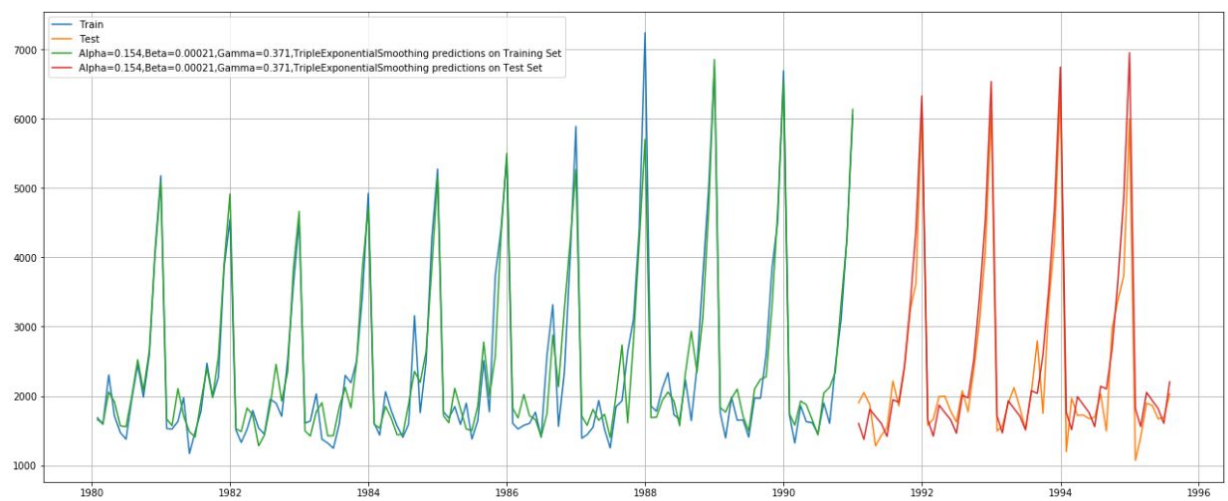
Double Exponential Smoothing Manual



Alpha=0.0,Beta=0.0,DoubleExponentialSmoothing	1460.954675

- After many iterations the model with the lowest RMSE has been applied
- At Alpha and Beta Level 0 the plot suggests no correct forecast has been made

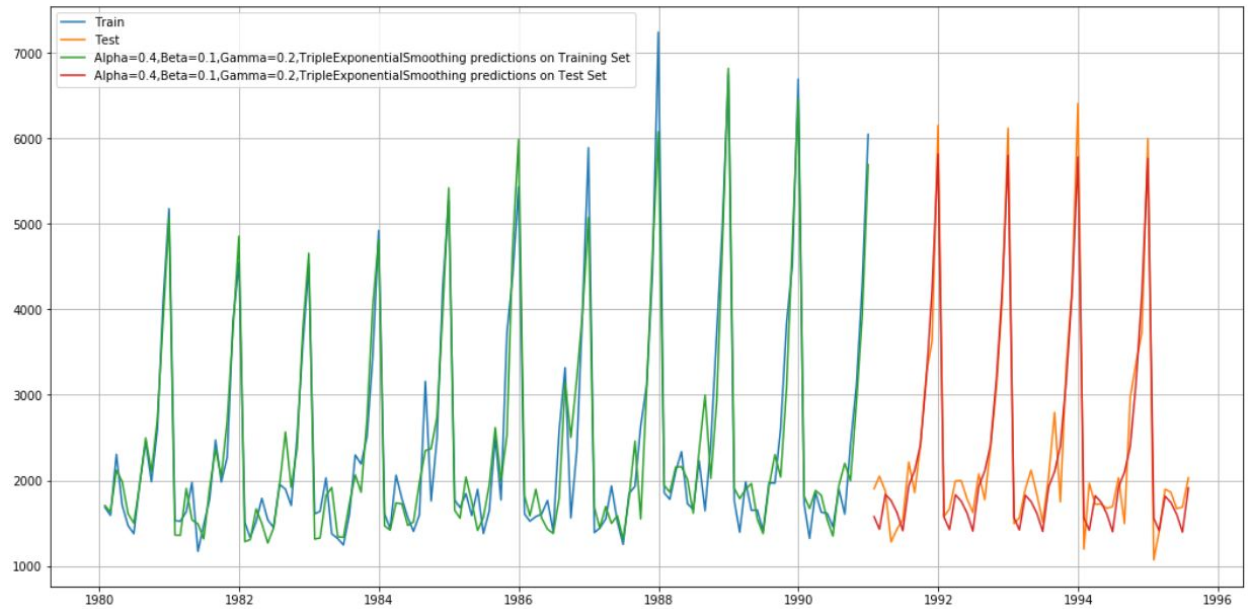
Triple Exponential smoothing Automated



Alpha=0.154,Beta=0.00021,Gamma=0.371,TripleExponentialSmoothing	384.203001

- Triple Exponential Smoothing has been applied using best parameters
- At Alpha Level 0.15, Beta Level 0.00021 and Gamma Level 0.37 the model shows good performance as it has the lowest RMSE of all
- The Plot shows the model has successfully captured the seasonality and it has performed the closest forecast of all models

Triple Exponential Smoothing Manual

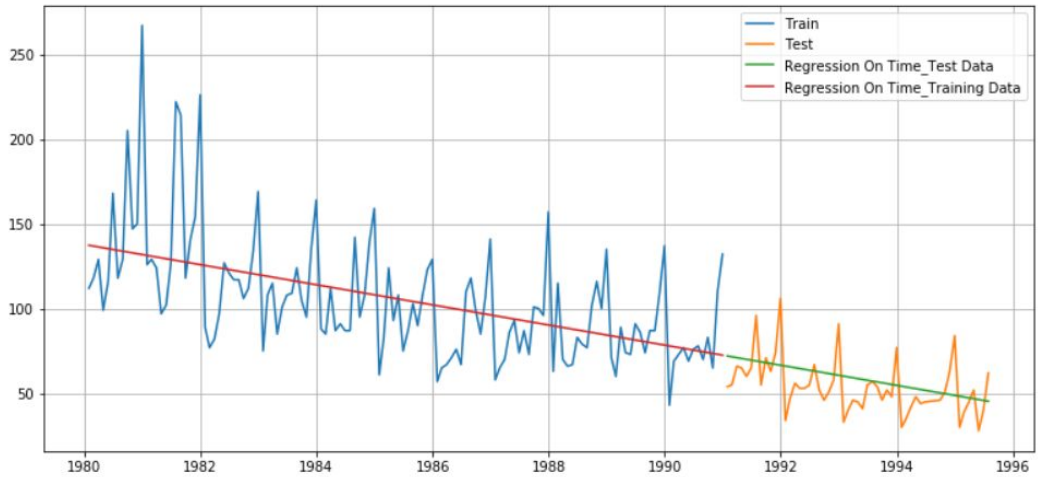


Alpha=0.4,Beta=0.1,Gamma=0.2,TripleExponentialSmoothing	312.211095

- After many iterations the best parameters have been chosen manually
- At Alpha level 0.4, Beta level 0.1 and Gamma level 0.2 the model has done better performance than the previous iteration based on its RMSE
- The plot suggests it has done better at forecasting than the previous iteration

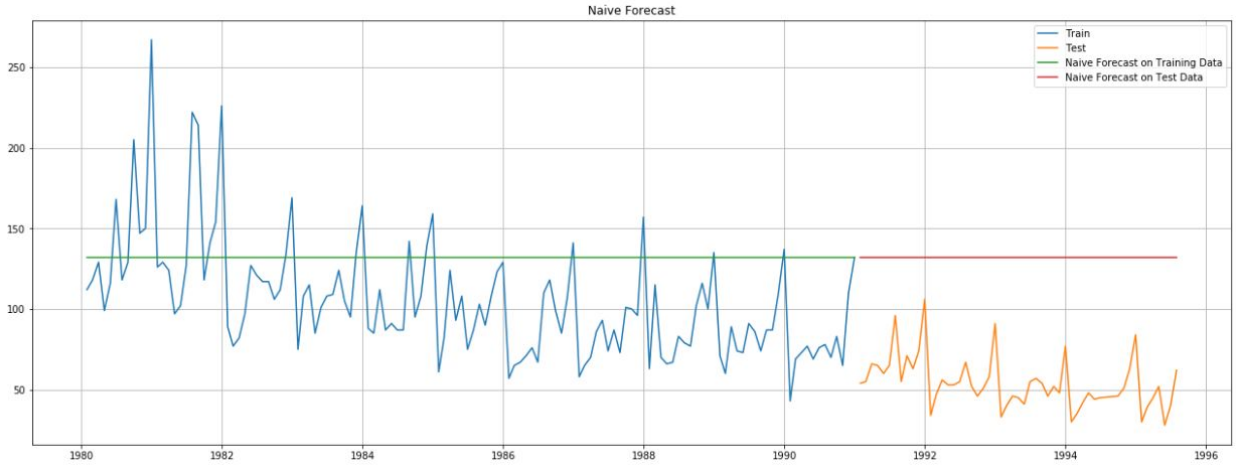
Rose

Linear Regression



- Linear Regression model has been fitted and applied on Test Data
- RMSE has been calculated at 15.27
- The Plot suggests the model fails at predicting the correct sales as it is not able to capture the seasonality of the data

Naive Model

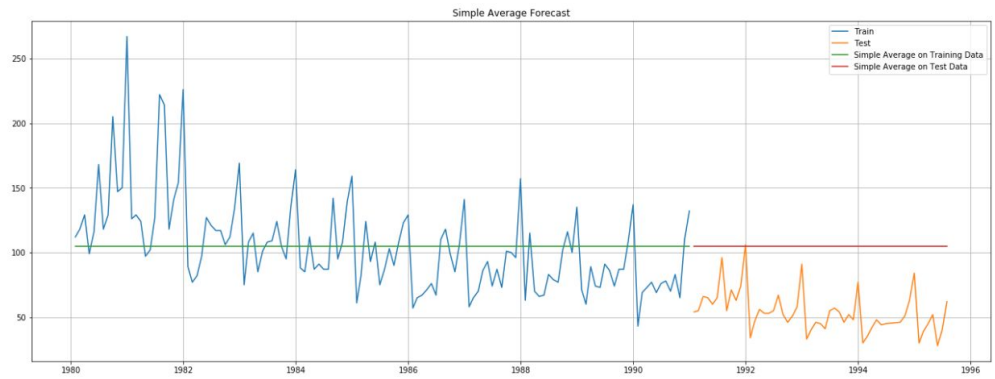


NaiveModel	79.718773

- Naive Model has been built and fitted over Test Data

- The RMSE of the model is higher than that of Linear Regression which means it performed poorly than Linear regression
- The Plot suggests it has failed to forecast the sales

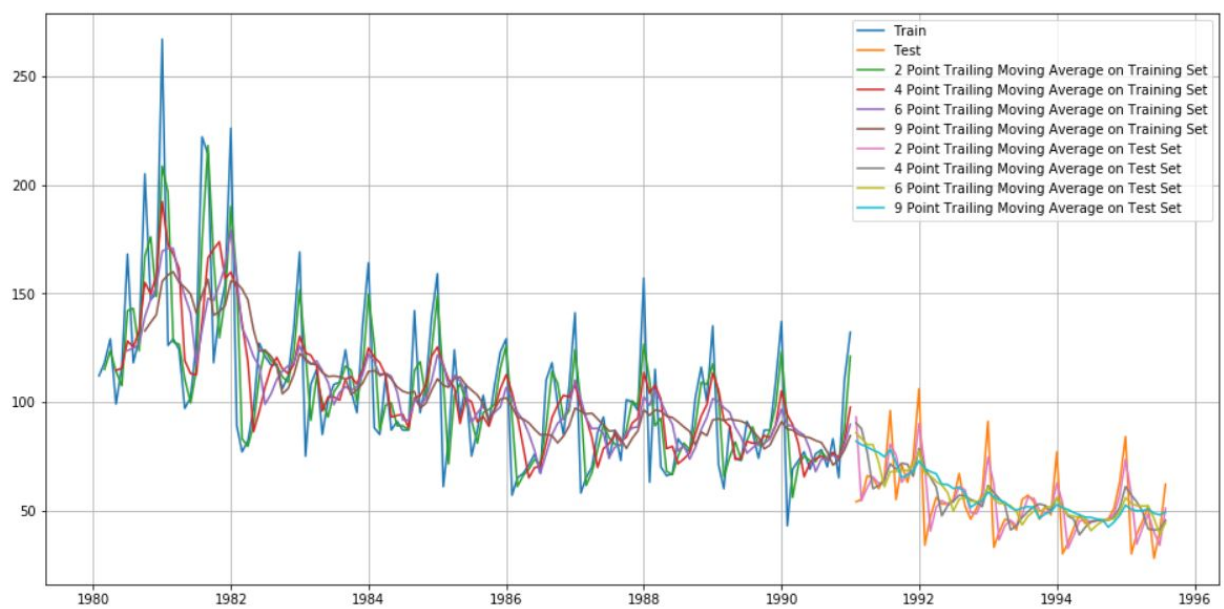
Simple Average



SimpleAverageModel	53.460570

- The RMSE of the Simple Average Model is lower than Naive model which suggests it has done better at decreasing its errors when compared to Naive
- However the plot suggests it too has failed at forecasting correct Sales

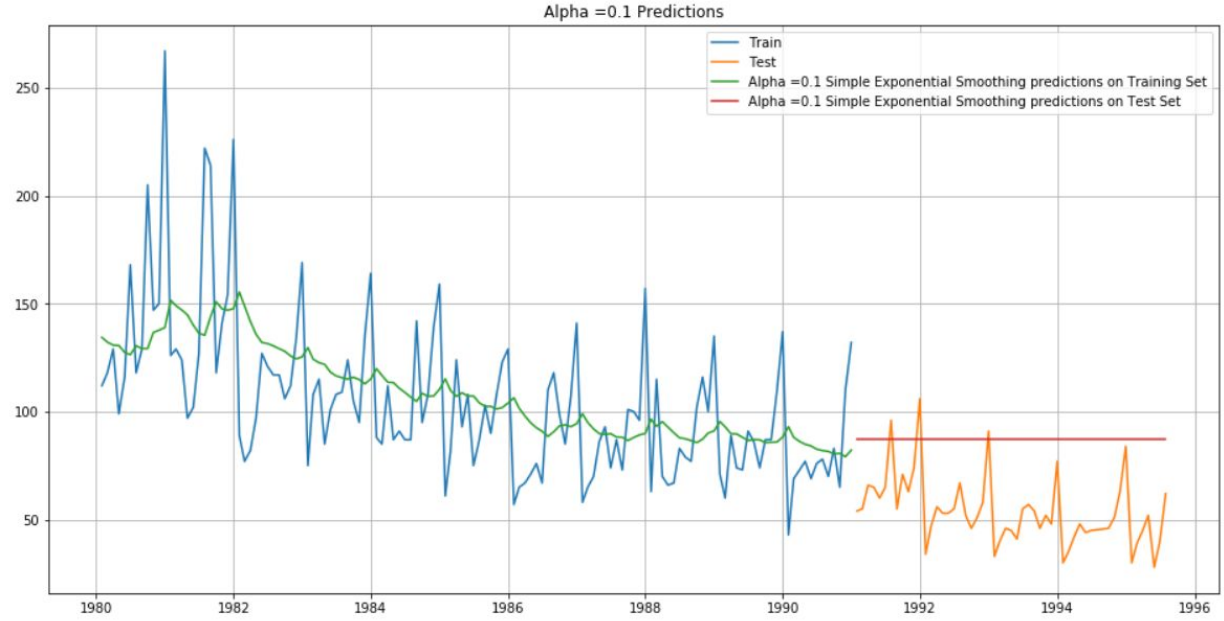
Moving Average



2pointTrailingMovingAverage	11.529278
4pointTrailingMovingAverage	14.451403
6pointTrailingMovingAverage	14.566327
9pointTrailingMovingAverage	14.727630

- Moving Average model has been applied at 2,4,6 and 9 Points
- The RMSE suggests that 2point Moving Average has done a better job at reducing its errors when compared with all other models
- The plot shows that 2point Moving Average comes significantly close to forecasting exact Sales of Rose Wine

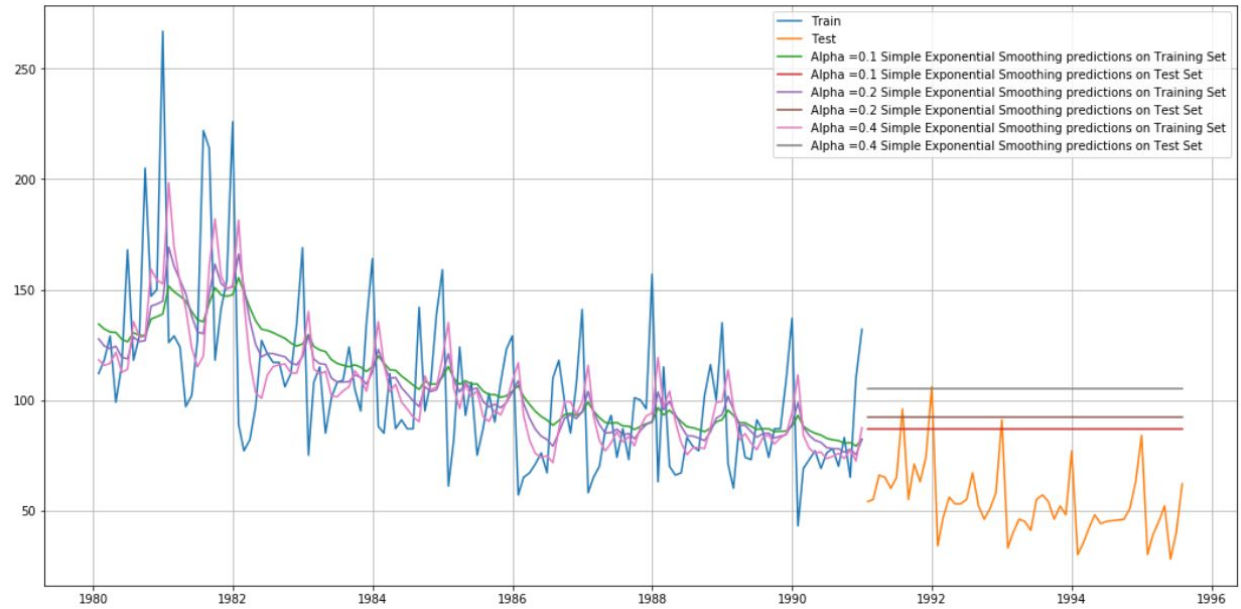
Simple Exponential smoothing Automated



Alpha=0.1,SimpleExponentialSmoothing	36.796242

- Simple Exponential Smoothing Model has been fitted using best parameters suggested by Statsmodels
- At Alpha Level 0.1 the model's Performance has done poorly according to RMSE
- The plot shows it fails to forecast on both Train and Test

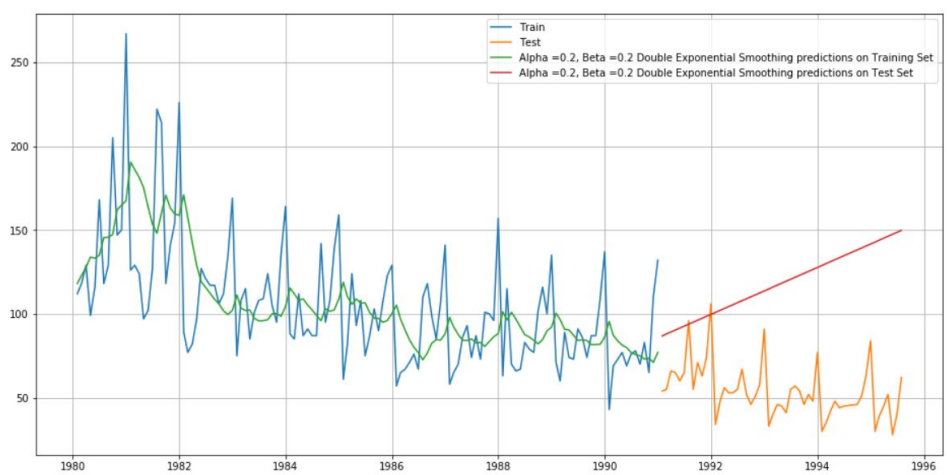
Simple Exponential Smoothing Manual



Alpha=0.2,SimpleExponentialSmoothing	41.361876
Alpha=0.4,SimpleExponentialSmoothing	53.767406

- After different iterations of the model it still fails in forecasting correct Sales
- Both at Alpha Level 2 and 4 the model is overfitted as it works on Train but fails on test Data as observed in the plot
- The RMSE of both iterations are high

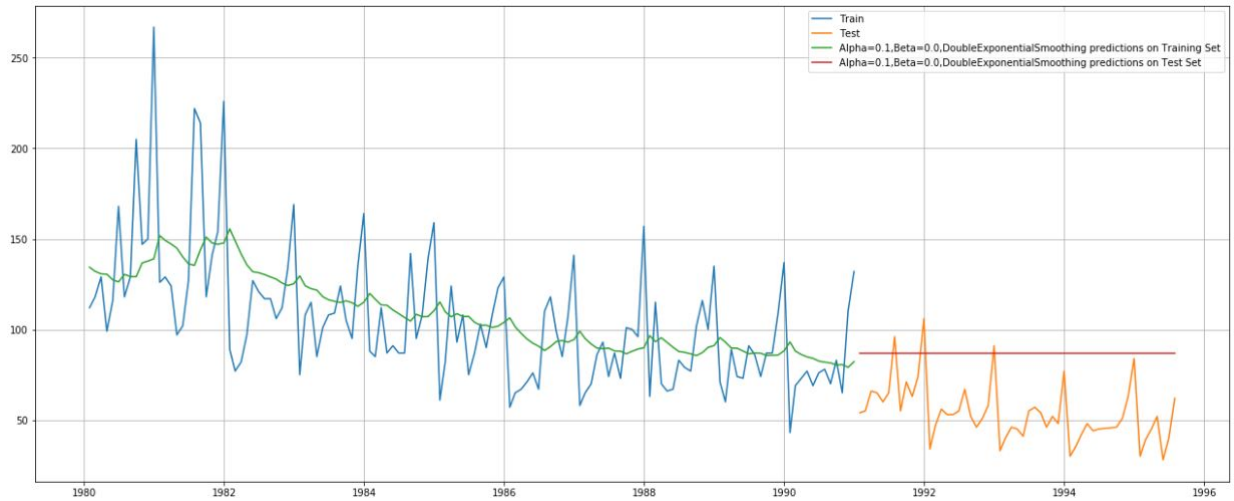
Double exponential smoothing automated



Alpha=0.2 Beta =0.2,DoubleExponentialSmoothing 70.572452

- Double exponential smoothing model has been applied using best parameters
- At Alpha Level 0.2 and and Beta level 0.2 the model's performance is similar to that of Naive Model based on RMSE

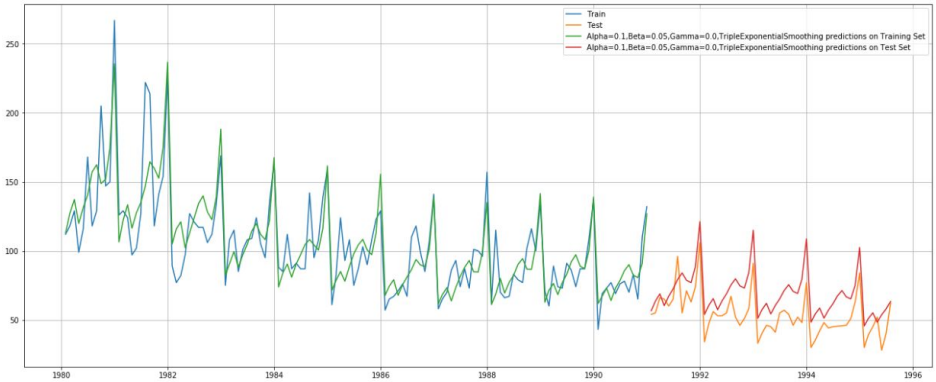
Double Exponential Smoothing manual



Alpha=0.1,Beta=0.0,DoubleExponentialSmoothing 36.828051

- After many iterations the model with the lowest RMSE has been applied
- At Alpha level 0.1 and Beta Level 0 the plot suggests no correct forecast has been made

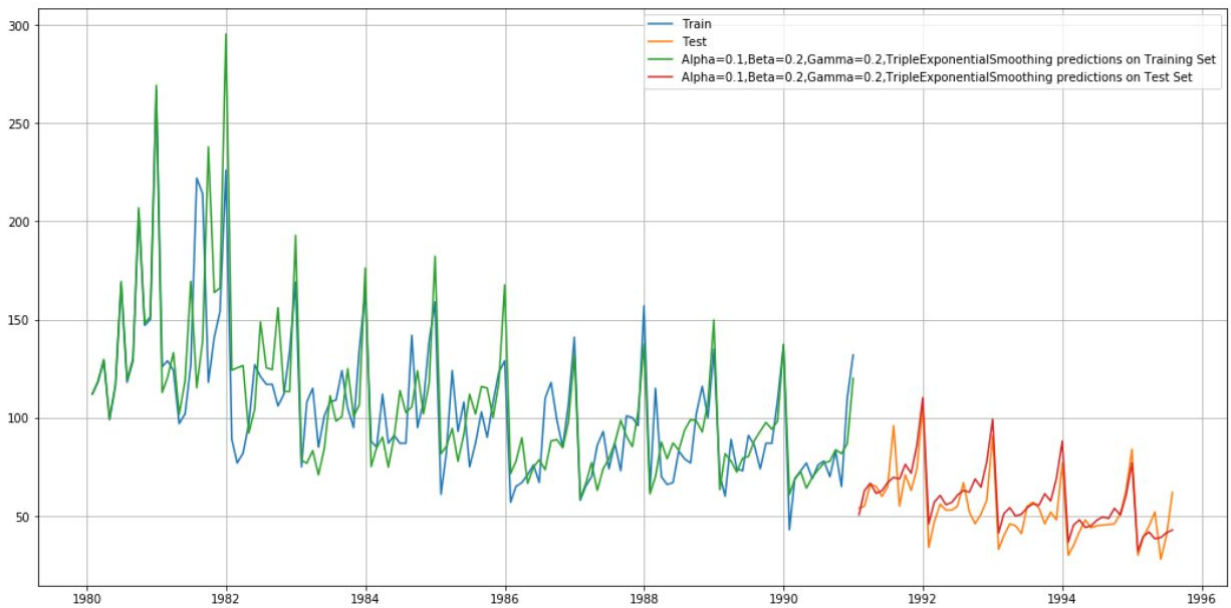
Triple Exponential Smoothing Automated



```
Alpha=0.106,Beta=0.05,Gamma=0.0,TripleExponentialSmoothing     17.369487
```

- Triple Exponential Smoothing has been applied using best parameters
- At Alpha Level 0.106, Beta Level 0.05 and Gamma Level 0 the model shows good performance as it has low RMSE
- The Plot shows the model has successfully captured the seasonality and it has performed the closest forecast of all models

Triple Exponential Smoothing Manual

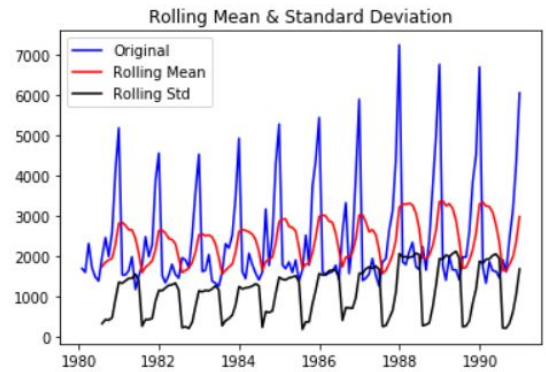



```
Alpha=0.1,Beta=0.2,Gamma=0.2,TripleExponentialSmoothing     9.640687
```

- After many iterations the best parameters have been chosen manually
- At Alpha level 0.1, Beta level 0.2 and Gamma level 0.2 the model has done a better performance than all models based on its RMSE
- The plot suggests it has done better at forecasting than the previous iteration

5. Check for the stationarity of the data on which the model is being built on using appropriate statistical tests and also mention the hypothesis for the statistical test. If the data is found to be non-stationary, take appropriate steps to make it stationary. Check the new data for stationarity and comment. Note: Stationarity should be checked at alpha = 0.05.

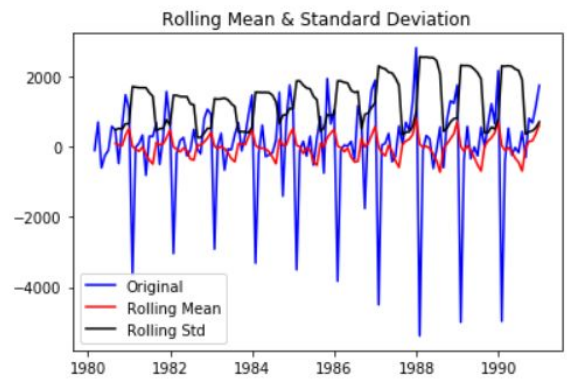
Sparkling




```
Results of Dickey-Fuller Test:
Test Statistic                  -1.208926
p-value                          0.669744
#Lags Used                      12.000000
Number of Observations Used    119.000000
Critical Value (1%)             -3.486535
Critical Value (5%)             -2.886151
Critical Value (10%)            -2.579896
dtype: float64
```

Data has been checked for Stationarity using the Dickey-Fuller test at an Alpha level of 0.05

- At 0.05 the P-value suggests the Data is non-stationary

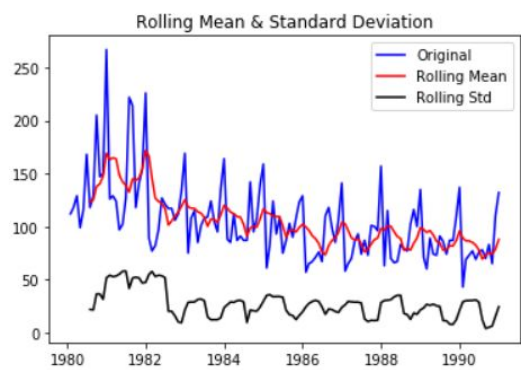



```
Results of Dickey-Fuller Test:
Test Statistic                  -8.005007e+00
p-value                          2.280104e-12
#Lags Used                       1.100000e+01
Number of Observations Used      1.190000e+02
Critical Value (1%)             -3.486535e+00
Critical Value (5%)             -2.886151e+00
Critical Value (10%)            -2.579896e+00
dtype: float64
```

The data is stationarized by taking a difference of order 1 and the Test is done again

- At 0.05 the P-value suggests the data is stationary after Differencing

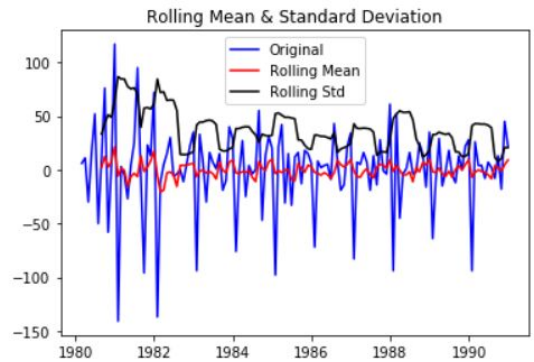
Rose




```
Results of Dickey-Fuller Test:
Test Statistic                  -2.164250
p-value                          0.219476
#Lags Used                      13.000000
Number of Observations Used    118.000000
Critical Value (1%)             -3.487022
Critical Value (5%)             -2.886363
Critical Value (10%)            -2.580009
dtype: float64
```

Data has been checked for Stationarity using the Dickey-Fuller test at an Alpha level of 0.05

- At 0.05 the P-value suggests the Data is non-stationary




```
Results of Dickey-Fuller Test:
Test Statistic                  -6.592372e+00
p-value                          7.061944e-09
#Lags Used                       1.200000e+01
Number of Observations Used      1.180000e+02
Critical Value (1%)             -3.487022e+00
Critical Value (5%)             -2.886363e+00
Critical Value (10%)            -2.580009e+00
dtype: float64
```

The data is stationarized by taking a difference of order 1 and the Test is done again

- At 0.05 the P-value suggests the data is stationary after Differencing

6. Build an automated version of the ARIMA/SARIMA model in which the parameters are selected using the lowest Akaike Information Criteria (AIC) on the training data and evaluate this model on the test data using RMSE.

Rose

SARIMA(0, 1, 2)(2, 0, 2,12)	26.928362

Sparkling

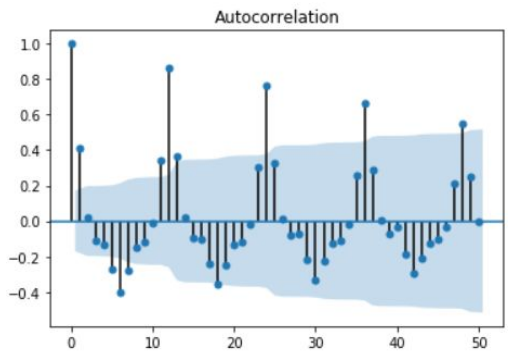
SARIMA(1, 1, 2)(0, 1, 2,12)	382.576693

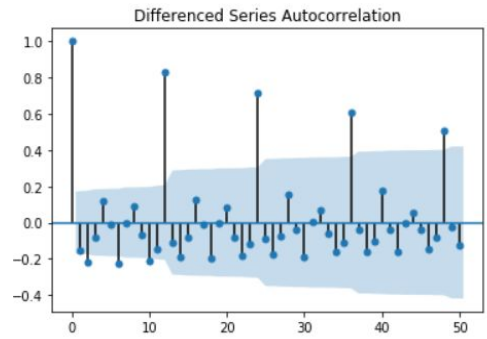
- Since The Data has seasonality SARIMA has been used on both datasets
- An automated version has been built using parameters with lowest AIC and applied on the Test Data with order 1 differencing
- Based on the RMSE the model has performed moderately on Sparkling but good on Rose

7. Build ARIMA/SARIMA models based on the cut-off points of ACF and PACF on the training data and evaluate this model on the test data using RMSE.

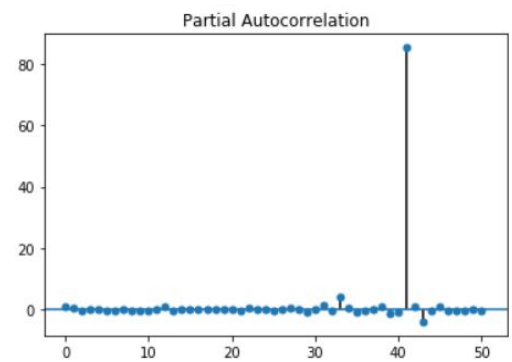
Sparkling

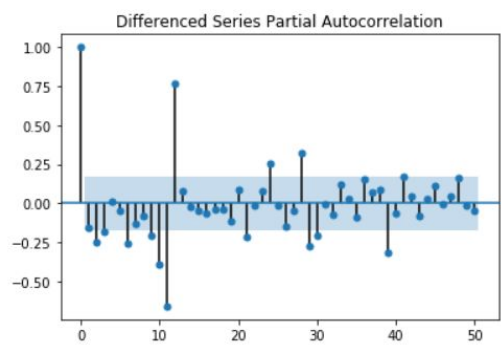
ACF Plot





PACF Plot



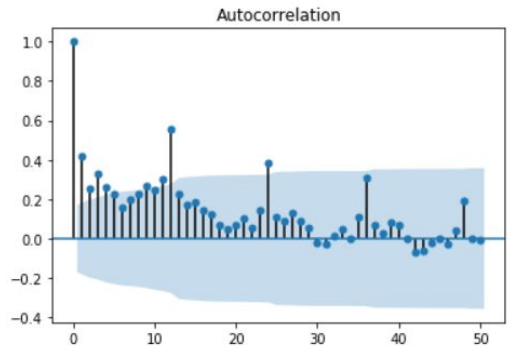


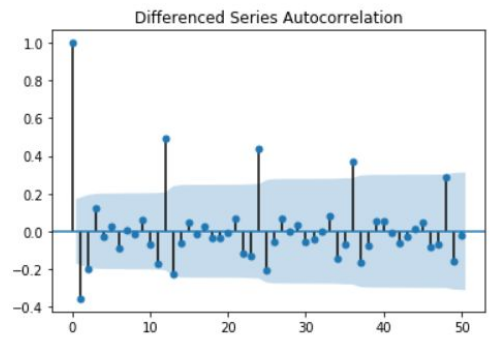
SARIMA(1, 1, 1)(1, 1, 1,12) 375.614011

- The parameters have been changed based off the cut off points of the ACF and PACF plots at order 1 differencing
- The model then model shows better performance than the previous model based on its RMSE

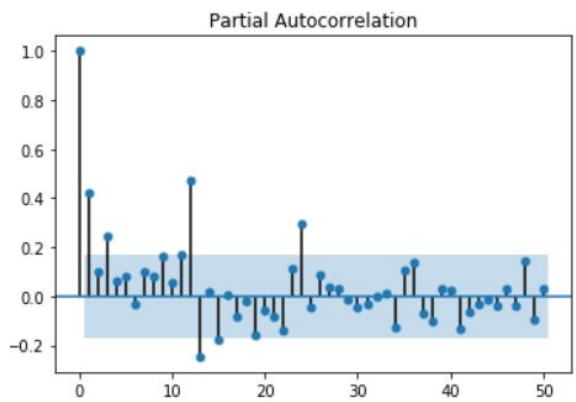
Rose

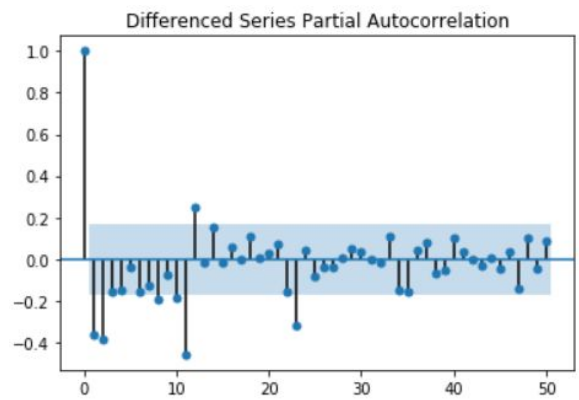
ACF Plot





PACF Plot





SARIMA(1, 1, 1)(0, 1, 1,12)	16.684996

- The parameters have been changed based off the cut off points of the ACF and PACF plots at order 1 differencing
- The model then model shows better performance than the previous model based on its RMSE

8. Build a table (create a data frame) with all the models built along with their corresponding parameters and the respective RMSE values on the test data.

- A DataFrame has been created with all the models built along with the respective RMSE of Test Data
- Triple Exponential Smoothing with manually chosen parameters shows the best performance at lowest RMSE for both Sparkling and Rose dataset

Sparkling

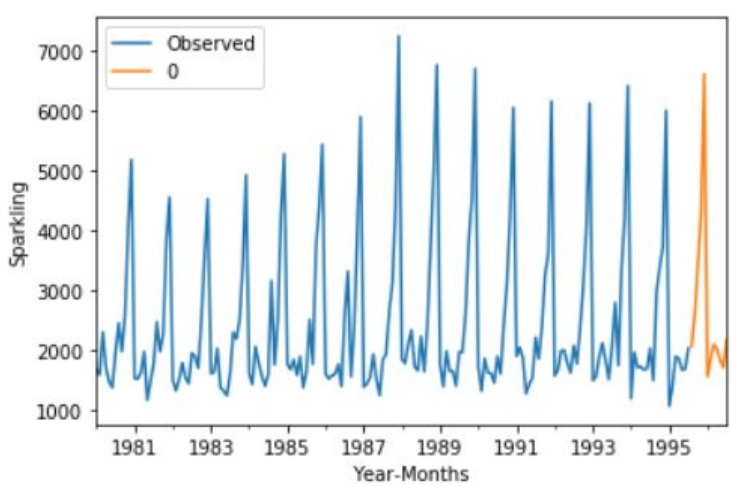
	Test RMSE
RegressionOnTime	1389.135175
NaiveModel	3864.279352
SimpleAverageModel	1275.081804
2pointTrailingMovingAverage	813.400684
4pointTrailingMovingAverage	1156.589694
6pointTrailingMovingAverage	1283.927428
9pointTrailingMovingAverage	1346.278315
Alpha=0,SimpleExponentialSmoothing	1275.081823
Alpha=0.1,SimpleExponentialSmoothing	1375.393526
Alpha=0.2,SimpleExponentialSmoothing	1595.206839
Alpha=0.6 Beta =0,DoubleExponentialSmoothing	3851.171500
Alpha=0.0,Beta=0.0,DoubleExponentialSmoothing	1460.954675
Alpha=0.154,Beta=0.00021,Gamma=0.371,TripleExponentialSmoothing	384.203001
Alpha=0.4,Beta=0.1,Gamma=0.2,TripleExponentialSmoothing	312.211095
SARIMA(1, 1, 2)(0, 1, 2,12)	382.576693
SARIMA(1, 1, 1)(1, 1, 1,12)	375.614011

Rose

	Test RMSE
RegressionOnTime	15.268955
NaiveModel	79.718773
SimpleAverageModel	53.460570
2pointTrailingMovingAverage	11.529278
4pointTrailingMovingAverage	14.451403
6pointTrailingMovingAverage	14.566327
9pointTrailingMovingAverage	14.727630
Alpha=0.1,SimpleExponentialSmoothing	36.796242
Alpha=0.2,SimpleExponentialSmoothing	41.361876
Alpha=0.4,SimpleExponentialSmoothing	53.767406
Alpha=0.2 Beta =0.2,DoubleExponentialSmoothing	70.572452
Alpha=0.1,Beta=0.0,DoubleExponentialSmoothing	36.828051
Alpha=0.106,Beta=0.05,Gamma=0.0,TripleExponentialSmoothing	17.369487
Alpha=0.1,Beta=0.2,Gamma=0.2,TripleExponentialSmoothing	9.640687
SARIMA(0, 1, 2)(2, 0, 2,12)	26.928362
SARIMA(1, 1, 1)(0, 1, 1,12)	16.684996

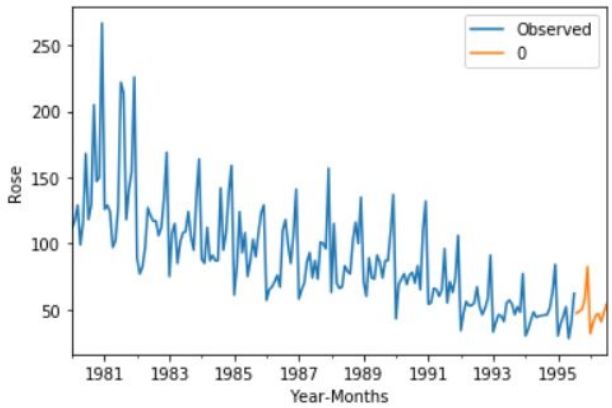
9. Based on the model-building exercise, build the most optimum model(s) on the complete data and predict 12 months into the future with appropriate confidence intervals/bands.

Sparkling



- Triple Exponential Smoothing has been built at Alpha Level 0.4, Beta level 0.1 and Gamma Level 0.2 on the whole data which has then been used to forecast the Sales for next 12 months
- The Plot shows forecasted Sales for next 12 Months with Seasonality

Rose



- Triple Exponential Smoothing has been built at Alpha Level 0.1, Beta level 0.2 and Gamma Level 0.2 on the whole data which has then been used to forecast the Sales for next 12 months
- The Plot shows forecasted Sales for next 12 Months with Seasonality

10. Comment on the model thus built and report your findings and suggest the measures that the company should be taking for future sales.

Sparkling

- The Company should increase Stock for October, November and December as these are holiday and festival months, and should expect high sales during these months
- The First half of the Year the company is recommended to expect moderate sales
- The forecast made on the whole Data has seasonality and thus shows when to expect high sales and when to expect low sales over the course of years
- The company should stock up accordingly in the upcoming year
- Since it is Multiplicative The company should expected a longer season than the previous one
- However since there is no Trend the company is recommended not to increase its yearly total quantity of Wine Supply

Rose

- The general Sales of Rose Wine is relatively and thus recommended to minimize stock
- During October, November and December the company can expect a slight increase in Sales
- The rest of the year moderate sales is to be expected
- The forecast made on the whole data shows seasonality will be short due to Multiplicative behaviour
- Thus it is recommended to expect a very short season for the upcoming year
- The data has a decreasing trend thus it is recommended that the stock of Rose Wine be decreased for the upcoming year